3. Common Distributions

This chapter introduces common continuous distributions.

The main distributions presented in this chapter are

- uniform
- normal
- exponential

Other continuous distribution to be briefly discussed include the Gamma, Chi-square, multivariate normal,…

First, two discrete distributions are reviewed.

The binomial distribution which can be approximated by other distributions.

The Poisson distribution with its properties that are useful in practical applications and which is related to the exponential distribution.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.1 Binomial and Poisson Distributions

3.1.1 Binomial distribution

A **binomial random variable,** with parameters n and p, has a probability mass function given by

$$P\{X = x\} = \binom{n}{x} p^x (1-p)^{n-x}$$

It is the probability that x successes will occur in n independent trials, each with a probability of success equal to p.

Its mean and variance are $\mathrm{E}[X] = np$ and $Var[X] = np(1-p)$.

Its moment generating function is $\phi(t) = M_X(t) = (1 - p + pe^t)^n$.

Note: Since $X = \sum_{i=1}^{n} X_i$, where $X_i = \begin{cases} 1 & \text{success}, p \\ 0 & \text{failure}, 1-p \end{cases}$,

$$M_X(t) = \left(M_{X_i}(t)\right)^n = (E[e^{tX_i}])^n = (1 - p + pe^t)^n$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.1.2 Poisson distribution

The probability mass function of a **Poisson random variable** with parameter λ is

$$P\{X = x\} = e^{-\lambda}\frac{\lambda^x}{x!}, \qquad x = 0,1,2, \ldots$$

Its moment generating function is

$$\phi(t) = M_X(t) = E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx}\ e^{-\lambda}\frac{\lambda^x}{x!} = e^{-\lambda}\sum_{x=0}^{\infty}\frac{(\lambda e^t)^x}{x!} = e^{-\lambda}\exp\{\lambda e^t\}$$
$$= \exp\{\lambda(e^t - 1)\}$$

We can find its mean and variance as follows:

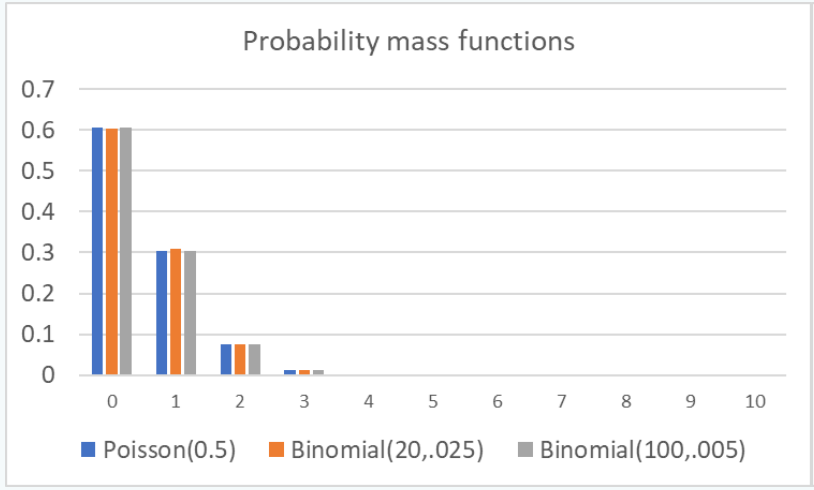
$$\mathrm{E}[X] = \phi'(0) = \lambda e^t \exp\{\lambda(e^t - 1)\}|_{t=0} = \lambda$$
$$Var[X] = \phi''(0) - \mathrm{E}[X]^2$$
$$= (\lambda e^t)^2 \exp\{\lambda(e^t - 1)\} + \lambda e^t \exp\{\lambda(e^t - 1)\}|_{t=0} - \lambda^2$$
$$= \lambda^2 + \lambda - \lambda^2 = \lambda$$

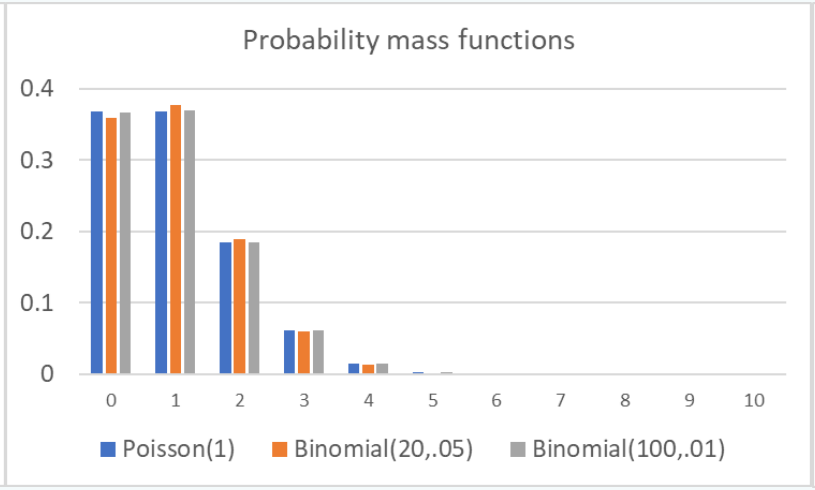
THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

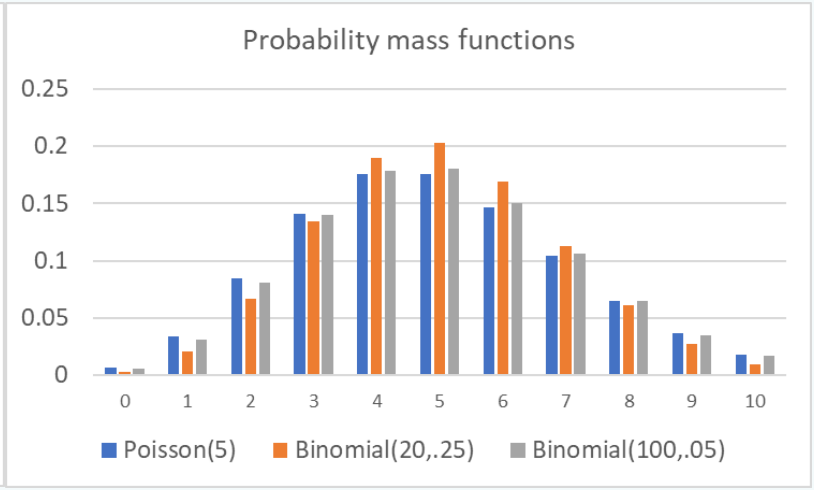
The Poisson distribution is a good approximation to

- # of misprints on a page of a book,
- # of people in a community living to 100,
- # of electronic components that fail on their first day of use,
- # of customers entering a store on a given day.

It is also a good approximation for a binomial distribution with *n* large and *p* small.







THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Reproductive property

Consider $X_1 \sim Po(\lambda_1)$ and $X_2 \sim Po(\lambda_2)$, two independent Poisson random variables.

Then, by the reproductive property of the Poisson distribution, their sum is also Poisson with $\lambda = \lambda_1 + \lambda_2$.

Proof:

$$\phi_{X_1+X_2}(t) = E\left[e^{t(X_1+X_2)}\right] = E\left[e^{t(X_1)}\right] E\left[e^{t(X_2)}\right] \qquad \text{Independence}$$

$$= \exp\{\lambda_1(e^t - 1)\} \exp\{\lambda_2(e^t - 1)\}$$

$$= \exp\{(\lambda_1 + \lambda_2)(e^t - 1)\}$$

$$= \exp\{\lambda(e^t - 1)\} \qquad \text{where } \lambda = \lambda_1 + \lambda_2$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: Consider 20,000 couples.

a) Approximate the probability that for at least one couple, both were born on February 28.

b) Approximate the probability that for at least one couple, both were born on the same day of the year.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Consider $N \sim Po(\lambda)$, a Poisson random variable with mean λ, and suppose that each event is **independently classified** as type i, $i=1,2,\ldots,r$ with probability p_i where $\sum_{i=1}^{r} p_i = 1$.

Then the numbers of events of type i have independent Poisson distributions with respective parameters λp_i ; i.e. $N_i \sim Po(\lambda p_i)$, $i=1,2,\ldots,r$.

Proof for two types of events: Here, $N = N_1 + N_2$

$$P\{N_1 = n, N_2 = m\} = P\{N_1 = n, N = n+m\} = P\{N_1 = n | N = n+m\}\, P\{N = n+m\}$$

Using the fact that $N_1 | (N = n+m) \sim Bin(n+m, p)$

$$P\{N_1 = n, N_2 = m\} = \binom{n+m}{n} p^n (1-p)^m \; e^{-\lambda} \frac{\lambda^{n+m}}{(n+m)!}$$

$$= \frac{(n+m)!}{n!m!} p^n (1-p)^m \; e^{-\lambda} \frac{\lambda^{n+m}}{(n+m)!}$$

$$= e^{-\lambda p} \frac{(\lambda p)^n}{n!} \; e^{-\lambda(1-p)} \frac{(\lambda(1-p))^m}{m!}$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

$$P\{N_1 = n\} = \sum_{m=0}^{\infty} P\{N_1 = n, N_2 = m\}$$

$$= e^{-\lambda p}\frac{(\lambda p)^n}{n!}\underbrace{\sum_{m=0}^{\infty} e^{-\lambda(1-p)}\frac{(\lambda(1-p))^m}{m!}}_{\text{pmf for } \mathrm{Po}(\lambda(1-p)) \Rightarrow \text{ sum } = 1}$$

$$= e^{-\lambda p}\frac{(\lambda p)^n}{n!} \quad \text{the pmf of } N_1 \sim Po(\lambda p).$$

Similarly, $P\{N_2 = m\} = e^{-\lambda(1-p)}\frac{(\lambda(1-p))^m}{m!}$ $N_2 \sim Po(\lambda(1-p))$

And $P\{N_1 = n, N_2 = m\} = P\{N_1 = n\}\, P\{N_2 = m\}$

$\Rightarrow N_1$ and N_2 are independent.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: The number of patients arriving at a regional hospital has a Poisson distribution with mean 20 per hour. Assume that 10% of the patients were involved in an accident.

Find the probability that two or more patients arriving at the hospital over a 2-hour period were involved in an accident.

What if the average number of patients is 15 the first hour and 25 the second hour?

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2 Continuous Distributions

3.2.1 Uniform distribution

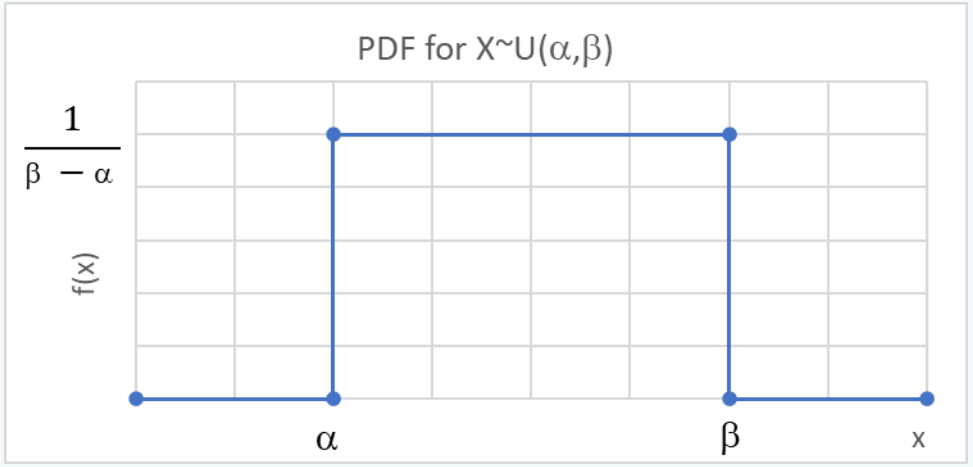
A random variable, X, is uniformly distributed over $[\alpha, \beta]$, $\mathrm{X} \sim \mathrm{U}[\alpha, \beta]$, if its pdf is

$$f(x) = \begin{cases} \dfrac{1}{\beta - \alpha} & \text{if } \alpha \le x \le \beta \\ 0 & \text{Otherwise} \end{cases}$$

It has a mean of $E[X] = \frac{\alpha+\beta}{2}$

and a variance of $Var[X] = \frac{(\beta-\alpha)^2}{12}$.

Its mgf is $\phi(t) = \begin{cases} \dfrac{e^{t\beta} - e^{t\alpha}}{t(\beta-\alpha)} & t \neq 0 \\ 1 & t = 0 \end{cases}$.



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example (see 5.4.c): The current in a semiconductor diode is often measured by the Shockley equation $I = I_0(e^{aV} - 1)$. Find $E[I]$ if $I_0 = 10^{-6}$, $a = 5$, and $V \sim U[1,3]$.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: Let X_1 and X_2 be independent U[0,1]. Find the cdf and pdf of $\mathrm{X} = X_1 + X_2$.

Note: The vector (X_1, X_2) has a two-dimensional uniform distribution on (0,1)x(0,1).

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2.2 Normal distribution

The probability density function of a **Normal random variable** with parameters μ and σ^2, $X \sim N(\mu, \sigma^2)$, is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \qquad -\infty < x < \infty. \qquad (*)$$

The pdf is symmetric about μ and has a maximum value of $\frac{1}{\sqrt{2\pi}\sigma} \approx \frac{0.4}{\sigma}$ when $x = \mu$.

It has two inflection points at $x = \mu \pm \sigma$.

It can be used as an approximation for a binomial distribution when *n* is large.

It is a good approximation to many empirical observations:

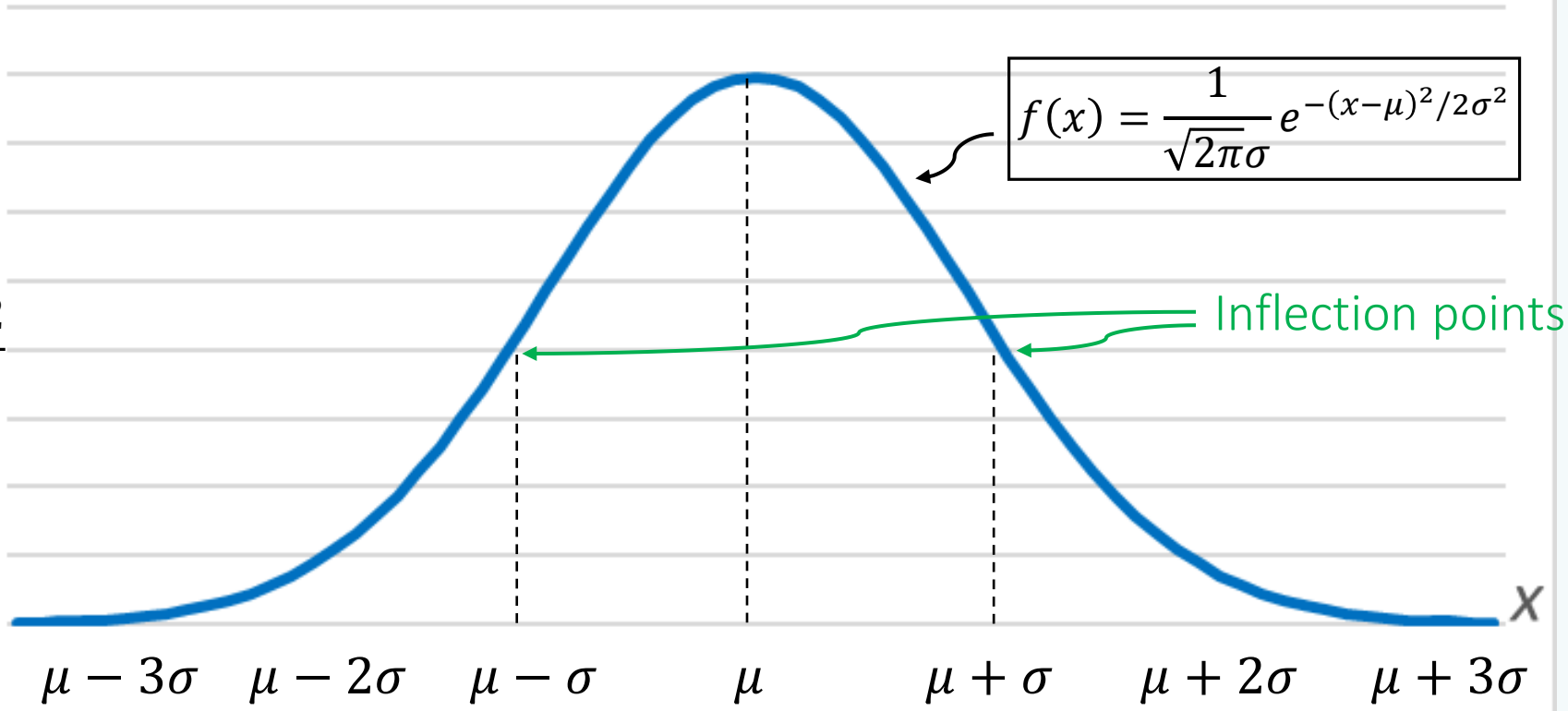
- Height of a person;
- Velocity of a molecule in a gas;
- Measurement errors of physical quantities.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Normal density function

$$\frac{1}{\sqrt{2\pi}\sigma} \approx \frac{0.399}{\sigma}$$

$$\frac{e^{-0.5}}{\sqrt{2\pi}\sigma} \approx \frac{0.242}{\sigma}$$



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

When $\mu = 0$ and $\sigma = 1$, it is called the **standard normal** distribution. Its pdf is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

To prove that the above pdf integrates to 1, let $I = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx$. Then

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2/2}\, dx \int_{-\infty}^{\infty} e^{-y^2/2}\, dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-y^2/2}\, e^{-x^2/2}\, dxdy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\, dxdy = \int_{0}^{2\pi}\int_{0}^{\infty} e^{-r^2/2}\, rdrd\theta \qquad \text{(Polar coordinates)}$$

$$= \int_{0}^{2\pi} -e^{-r^2/2}\Big|_0^{\infty} d\theta = \int_{0}^{2\pi} 1\, d\theta = 2\pi$$

Finally, $\int_{-\infty}^{\infty} f(z)\, dz = \frac{1}{\sqrt{2\pi}} I = 1$.

The expected value of the standard normal is

$$E[Z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2}\, dz = \frac{1}{\sqrt{2\pi}} \left(-e^{-z^2/2}\right)\Big|_{-\infty}^{\infty} = 0$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

The variance of Z is

$$Var(Z) = E[Z^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz \qquad \text{since } E[Z] = 0$$

Integrating by parts, $u = z$ $\quad dv = ze^{-z^2/2}\, dz$

$$du = dz \qquad v = -e^{-z^2/2}$$

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz = \underbrace{\left(-ze^{-z^2/2}\right)\Big|_{-\infty}^{\infty}}_{0} + \int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \int_{-\infty}^{\infty} e^{-z^2/2}\, dz$$

which gives

$$Var(Z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 1$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

The moment generating function of Z is

$$\phi(t) = E[e^{tZ}] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{tz}\ e^{-z^2/2}\,dz = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\ e^{-(z^2-2tz+t^2-t^2)/2}\,dz$$

$$= \frac{1}{\sqrt{2\pi}} e^{t^2/2}\int_{-\infty}^{\infty}\ e^{-(z-t)^2/2}\,dz$$

$$= e^{t^2/2}\underbrace{\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\ e^{-(y)^2/2}\,dy}_{1} \qquad \text{where } \mathrm{y} = z - t$$

$$\boldsymbol{\phi_Z(t) = e^{t^2/2}}$$

The properties of Z can be used to derive those of X with pdf given by equation (*).

Consider $X = \mu + \sigma Z$ with $\sigma > 0$. Then,

$$F_X(x) = P(X \le x) = P\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right) = F_Z\left(\frac{x-\mu}{\sigma}\right)$$

And $f_X(x) = \frac{d}{dx}F_X(x) = \frac{d}{dx}F_Z\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma}f_Z\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$ which is (*)

So, X is a normal random variable with $X \sim N(\mu, \sigma^2)$.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Using the properties of Z we can derive those of $X = \mu + \sigma Z$:

$X \sim N(\mu, \sigma^2)$

$E[X] = E[\mu + \sigma Z] = \mu + \sigma E[Z] = \mu$

$Var[X] = Var[\mu + \sigma Z] = \sigma^2 Var[Z] = \sigma^2$

and

$$\boldsymbol{\phi_X(t)} = E[e^{tX}] = E\left[e^{t(\mu+\sigma Z)}\right] = e^{t\mu}\, E\left[e^{(t\sigma)Z}\right] = e^{t\mu}\, \phi_Z(t\sigma) = e^{t\mu} e^{(t\sigma)^2/2}$$

$$= \boldsymbol{e^{\mu t + \sigma^2 t^2/2}}$$

Similar derivations would show that a linear transformation of a normal variable is also a normal variable.

If $X \sim N(\mu, \sigma^2)$, for any a and b, $b \neq 0$, $Y = a + bX$ is normal and

$Y \sim N(a + b\mu, b^2\sigma^2)$.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

The cumulative distribution function of a standard normal, denoted $\Phi(z)$, is

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-y^2/2}\ dy; \quad -\infty < z < \infty$$

There is no closed form for the cdf $\Phi(z)$.

There exist very good approximations and tabulated values of $\Phi(z)$ are widely available. Table A.1 of the Appendix in the textbook tabulates those values for nonnegative values of z.

Most software have a built-in function to calculate values for $\Phi(z)$ for any z. For example, in R the function ***pnorm***$(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\sigma})$ returns $\Phi\left(\frac{x-\mu}{\sigma}\right) = \mathrm{P}(\mathrm{X} \leq x)$ where $X \sim N(\mu, \sigma^2)$. In Excel, the function NORM.DIST can be used to return the pdf or the cdf of a normal distribution.

Note: By symmetry, $P(Z < -z) = P(Z > z)$ => $\Phi(-z) = 1 - \Phi(z)$.

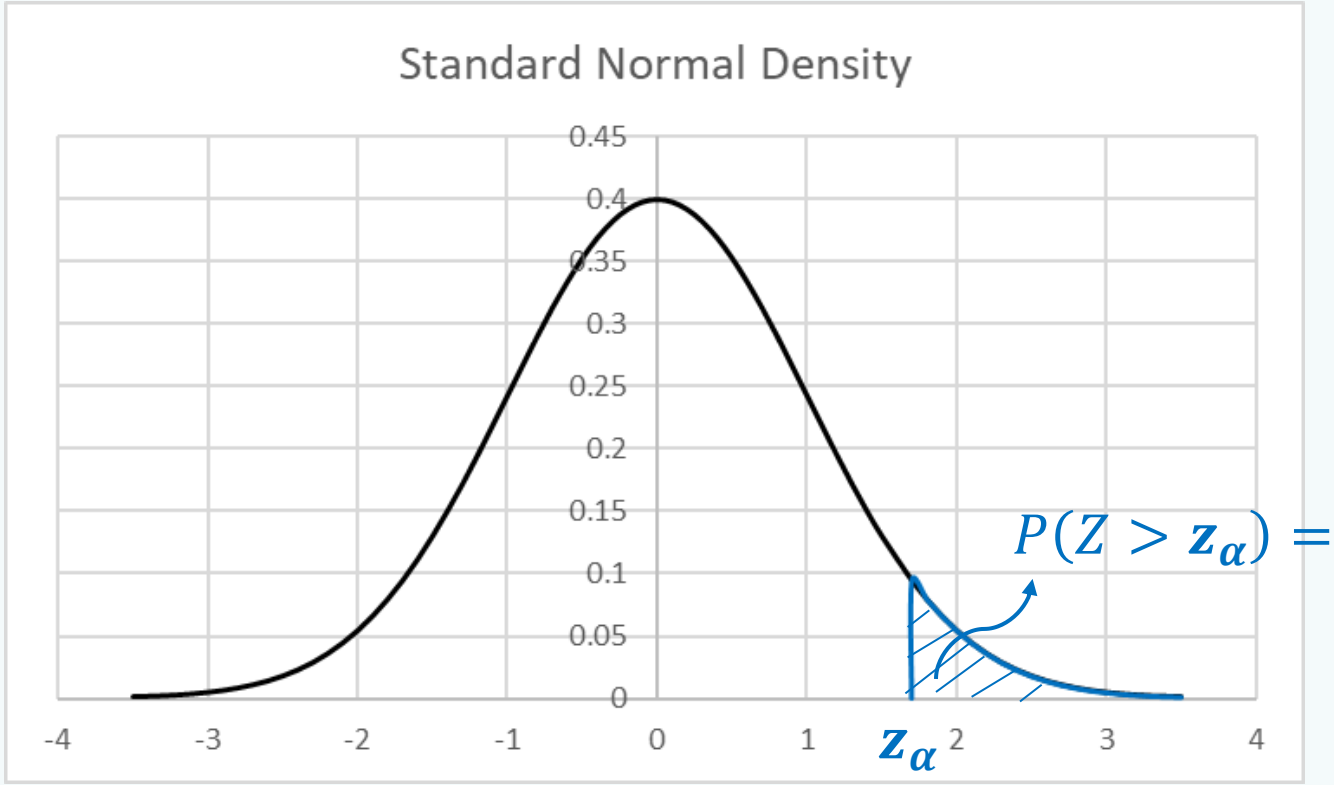
THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Percentiles of the standard normal distribution.

For $0 < \alpha < 1$, the $100(1-\alpha)$ percentile of the standard normal is z_α such that

$$P(Z > z_\alpha) = 1 - \Phi(z_\alpha) = \alpha$$

α	z_α
0.25	0.674
0.2	0.842
0.1	1.282
0.05	1.645
0.025	1.960
0.01	2.326



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example 1: (Example 5.5.b)

A binary message is transmitted by wire. The data sent is subject to a channel noise. Assume the value sent is x= -2 if the message is “0” ; and x=2 if the message is “1”.

The value received is R=x+N where N~N(0,1). The message is decoded as

“1” if R≥.5 and “0” if R<.5.

Find the probability of the two types of errors, i.e. message “0” received as “1”; and message “1” received as “0”.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example 2: (Example 5.5.c)

The power W dissipated in a resistor is proportional to the square of the voltage V.

$W = rV^2$ where r is a constant. If $r = 3$ and $V \sim N(6,1)$, find

a) $\mathrm{E}[W]$

b) $\mathrm{P}(W > 120)$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Sums of independent Normal random variables

The sum of independent normal variables is also normal.

Let $X_i \sim N(\mu_i, \sigma_i^2),\ i = 1,2,..,n.$ The mgf of their sum is

$$\phi_{\sum_{i=1}^n X_i}(t) = \prod_{i=1}^n \phi_{X_i}(t) = \prod_{i=1}^n e^{\mu_i \cdot t + \sigma_i^2 t^2/2} = e^{t\sum_{i=1}^n \mu_i + (\frac{t^2}{2})\sum_{i=1}^n \sigma_i^2}$$

$$= \boldsymbol{e^{\mu t + \sigma^2 t^2/2}} \quad \text{where } \mu = \sum_{i=1}^n \mu_i \text{ and } \sigma^2 = \sum_{i=1}^n \sigma_i^2$$

Since this is the mgf of a normal,

$$\sum_{i=1}^n X_i \sim N(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)$$

Note: The sum of linear combinations of normal variables is also normal.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: (Example 5.5.d)

The yearly precipitation (inches) in LA is $N(12.08, 3.1^2)$. Yearly precipitation totals are independent.

a) Find the probability that the total precipitation over the next 2 years will exceed 25.

b) Find the probability that next year's precipitation will exceed that on the following year by more than 3 inches.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2.3 Exponential distribution

The probability density function of an **Exponential random variable** with parameter λ, $X \sim Exp(\lambda)$, is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Its cumulative distribution is

$$F(x) = \int_0^x \lambda e^{-\lambda y}\, dy = 1 - e^{-\lambda x}, \qquad x \geq 0$$

In practice, it often approximates the time until a specific event occurs. For example,

-Time until an earthquake;

-Time until the next phone call;

-Time until the next client enters a store.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Moments of an Exponential variable

The mean and variance of an exponentially distributed variable have been calculated in some earlier examples (Chapter 2). We now derive its mgf and use it to find its moments.

The mgf of $X \sim Exp(\lambda)$ is

$$\phi(\mathrm{t}) = \mathrm{E}[e^{tX}] = \int_0^\infty e^{tx}\, \lambda e^{-\lambda x}\, dx = \lambda \int_0^\infty e^{-(\lambda - t)x}\, dx = \frac{\lambda}{\lambda - t}; \quad t < \lambda$$

So, $E[X] = \phi'(0) = \frac{\lambda}{(\lambda - t)^2}\Big| t = 0 = \frac{1}{\lambda}$

$$E[X^2] = \phi''(0) = \frac{2\lambda}{(\lambda - t)^3}\Big| t = 0 = \frac{2}{\lambda^2}$$

and

$$Var[X] = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Memoryless property of the Exponential distribution

Mathematically, X is memoryless if

$$P(X > s + t | X > t) = P(X > s), \qquad \text{for all } s, t \geq 0$$

For $X \sim Exp(\lambda)$,

$$P(X > s + t | X > t) = \frac{P(X>s+t, X>t)}{P(X>t)} = \frac{P(X>s+t)}{P(X>t)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = P(X > s)$$

So, an exponentially distributed random variable is memoryless.

An interpretation of this property is that the distribution of the length of time until the next event is independent of how long ago the last event occurred.

The memoryless condition is equivalent to

$$P(X > s + t) = P(X > t)\, P(X > s), \qquad \text{for all } s, t \geq 0$$

Note: Exponential variables are the only memoryless variables.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example 1:

The length of time it takes a clerk to serve a client has an exponential distribution with mean 2 minutes.

a) Find the probability that the clerk will need more than 5 minutes to serve a client.

b) Find the average time remaining to serve a client who has already been at the counter for 3 minutes.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example 2: (See example 5.6.b)

The time a machine functions before breaking down is exponential with parameter λ.

Three of those machine are available and two are required on a given job. Assume machine A and B are used first, and machine C is used to replace whichever of A or B breaks down first.

Find the probability that machine C is still operable when the job has to be stopped.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Minimum of independent Exponential variables

Consider n independent random variables with exponential distributions, $X_i \sim Exp(\lambda_i)$, i=1,2,…,n. Then, the minimum of these n variables is also exponential with parameter equal to the sum of their n parameters.

The complement of the cdf (i.e. the survival function), is

$$P(\min(X_1, X_2, \dots, X_n) > x) = P(X_1 > x, X_2 > x, \dots, X_n > x)$$
$$= \prod_{i=1}^{n} P(X_i > x) \quad \text{by independence}$$
$$= \prod_{i=1}^{n} e^{-\lambda_i x} = e^{-(\sum_{i=1}^{n} \lambda_i)x}$$

So, $\boldsymbol{min(X_1, X_2, \dots, X_n) \sim Exp(\sum_{i=1}^{n} \lambda_i)}$.

Interpretation: Assume n independent events will occur at future times that are exponentially distributed. Then, the time of the first occurrence will be exponentially distributed with parameter equal to the sum of the n parameters.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: A system functions if its 10 components are working. The lifetimes of 6 of the components are exponentially distributed with mean 100 hours. The lifetimes of the other 4 are exponentially distributed with mean 120 hours. The 10 lifetimes are independent.

a) Find the average time the system will function.

b) Find the probability that the system will function for at least 80 hours.

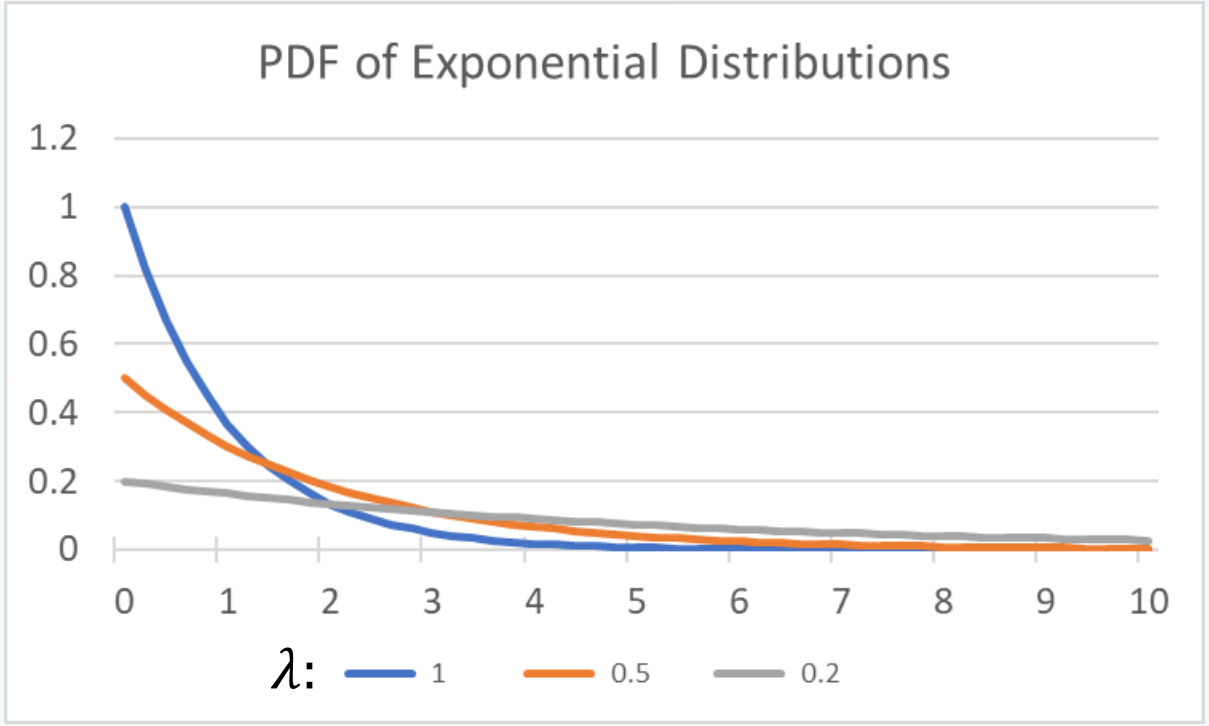
THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Scaling an Exponential random variable:

If $X \sim Exp(\lambda)$, then $cX \sim Exp(\lambda/c)$, for $c > 0$.

Note: λ is called the rate of the exponential distribution.

Illustrations of the exponential distribution (pdf)



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2.4 Poisson process and interarrival times

The number of events in *[0,t]*, *N(t)*, is a **Poisson process with rate $\lambda > 0$** , if

i) $N(0) = 0$; Process begins at 0

ii) $N(t_2) - N(t_1)$ and $N(t_4) - N(t_3)$ are independent for any $t_1 < t_2 < t_3 < t_4$;

Independent increments

iii) $N(t)$ and $N(t+s) - N(s)$ have identical distributions, *s>0;* Stationary increments

iv) $\lim_{h\to 0} \frac{P\{N(h)=1\}}{h} = \lambda$ and $\lim_{h\to 0} \frac{P\{N(h)\geq 2\}}{h} = 0$;

$P(1\ event\ in\ [0,h]) \approx \lambda h$ for h small

$P(2+\ events\ in\ [0,h]) \approx 0$

We can show that $\boldsymbol{N(t) \sim Po(\lambda t)}$.

So, $\mathrm{P}\{N(t) = k\} = e^{-\lambda t}\ \frac{(\lambda t)^k}{k!}$ $k = 0,1,2,\ldots$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

The (arrival) time of the first event of a Poisson process, X_1, has an exponential distribution with mean $1/\lambda$.

$$\mathrm{P}(X_1 > \mathrm{t}) = \mathrm{P}\{\mathrm{N}(\mathrm{t}) = 0\} = e^{-\lambda t}$$

Time of 1st event >t => no event in [0,t]

So, $X_1 \sim Exp(\lambda)$.

The elapsed time between event *n-1* and event *n*, X_n, has an exponential distribution.

$$\mathrm{P}(X_n > \mathrm{t} | X_{n-1} = s) = \mathrm{P}\{\mathrm{N}(\mathrm{t}+\mathrm{s}) - N(s) = 0 | N(s) = \mathrm{n}-1\} \qquad s < t$$

$$= \mathrm{P}\{\mathrm{N}(\mathrm{t}+\mathrm{s}) - N(s) = 0\} = e^{-\lambda t}$$

Independent increments

So, $X_n \sim Exp(\lambda)$.

For a Poisson process with rate λ, the sequence of interarrival times $\{X_1, X_2, \ldots\}$ is a sequence of independent exponential random variables with mean $1/\lambda$.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: The number of customers arriving at a store follows a Poisson process with rate 10 per hour. The store opens at 9:00 with no one waiting at the door.

a) Find the probability that exactly 4 customers arrive between 9:00 and 9:15.

b) Find the expected time of arrival of the 4th customer.

c) Find the probability that the first customer arrives after 9:15 given that it is 9:05 and no customer has arrived yet.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: The number of customers arriving at a store follows a Poisson process with rate 10 per hour. The store opens at 9:00 with no one waiting at the door.

d) You are given that the 5th customer arrived at 9:30.

Find the probability that the 8th customer will arrive after 10:00.

e) Find the expected value and standard deviation of the number of customers arriving before 10:00 (since 9:00).

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2.5 Other continuous distributions

Gamma Distribution

$X \sim Gamma(\alpha, \lambda)$ or $X \sim \Gamma(\alpha, \lambda)$ has a pdf given by

$$f(x) = \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \quad \text{for } x \geq 0.$$

In R: $dgamma(x, \alpha, \lambda)$

In Excel: $GAMMA.DIST(x, \alpha, \frac{1}{\lambda}, 0)$

The gamma function is $\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy$.

For integers, $\Gamma(n) = (n-1)! \quad n = 1,2, \ldots$

Its moment generating function is $\phi(\mathrm{t}) = (\frac{\lambda}{\lambda - t})^\alpha; \quad t < \lambda$

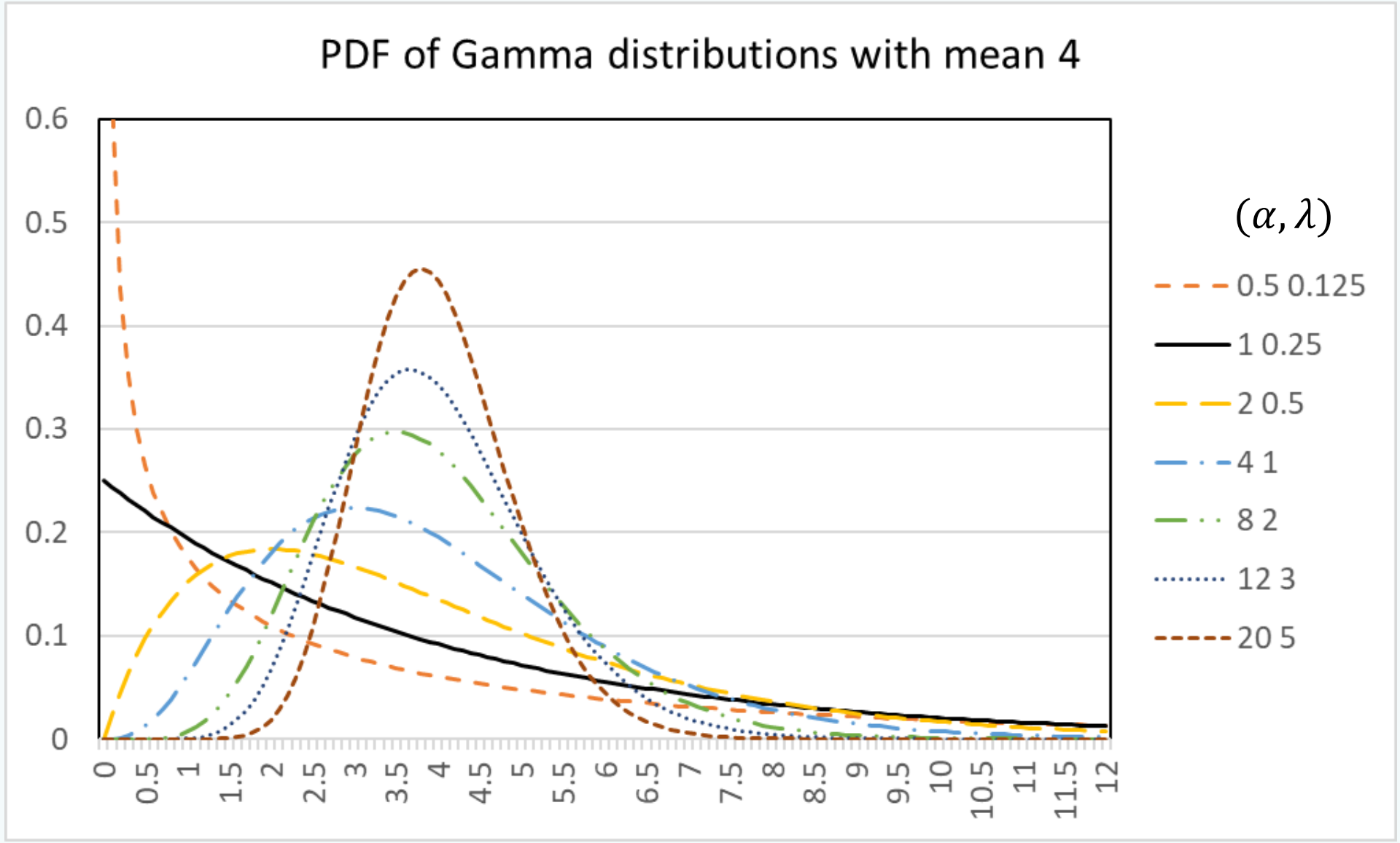
Its moments are $\mathrm{E}[X] = \frac{\alpha}{\lambda}$ and $\mathrm{Var}[X] = \frac{\alpha}{\lambda^2}$

Properties: i) With $\alpha = 1$, $\Gamma(1, \lambda)$ reduces to the exponential distribution, $X \sim Exp(\lambda)$

ii) For independent $X_i \sim \Gamma(\alpha_i, \lambda)$, i=1,2,…,n, we have $\sum_{i=1}^n X_i \sim \Gamma(\sum_{i=1}^n \alpha_i, \lambda)$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Illustrations of the Gamma distribution



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Chi-square Distribution

The chi-square distribution with n degrees of freedom, $X \sim \chi^2_n$, has a pdf given by

$$f(x) = \frac{0.5\, e^{-x/2}\, (x/2)^{(n/2)-1}}{\Gamma(n/2)} \quad \text{for } x \geq 0.$$

In R: $dchisq(x, n)$
In Excel: $CHISQ.DIST(x, \mathrm{n}, 0)$

Its moment generating function is $\phi(\mathrm{t}) = (1-2t)^{-n/2}; \quad t < 1/2$

Its moments are $\mathrm{E}[X] = n$ and $\mathrm{Var}[X] = 2n$

Properties:

i) The chi-square is a special case of the Gamma distribution where $\alpha = \frac{n}{2}$ and $\lambda = \frac{1}{2}$.

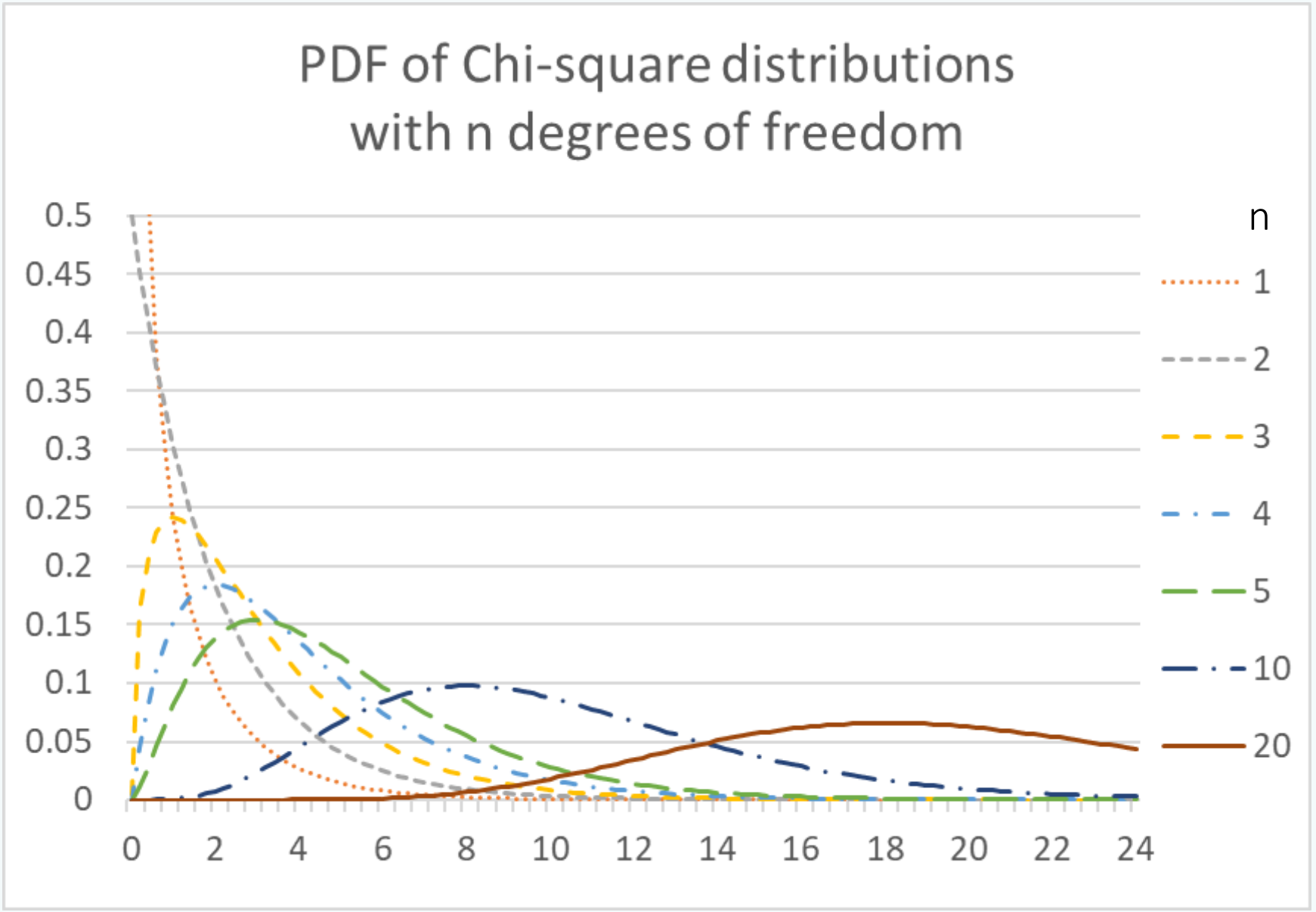
ii) $X = Z_1^2 + Z_2^2 + \cdots + Z_n^2$, where Z_i is a standard normal, has a χ^2_n distribution.

iii) When $n = 2$, $X \sim \chi^2_2$ is an exponential with mean 2, $X \sim Exp(\frac{1}{2})$

iv) For independent $X_i \sim \chi^2_{n_i}$, i=1,2,…,n, we have $\sum_{i=1}^n X_i \sim \chi^2_{\sum_{i=1}^n n_i}$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Illustrations of the chi-square distribution



THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Example: (see 5.8.c)

A target in 3-dimensional space is being located. The measurement errors (for each dimension) are independent N(0,4m^2). Find the probability that the distance between the chosen point and the target exceeds 3 meters.

The distance D is such that $D^2 = X_1^2 + X_2^2 + X_3^2$ where X_i is the error in dimension i.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

3.2.6 Multivariate Normal distribution (Multinormal)

A random vector $\boldsymbol{X}$ has a p-dimensional multivariate (or multinormal) distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, $\boldsymbol{X \sim N_p(\mu, \Sigma)}$, if its pdf is

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2}\,|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})'\,\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}$$

where

$\boldsymbol{x}$ is a (px1) vector

$\boldsymbol{\mu}$ is a (px1) vector with $E[X_i] = \mu_i$

$\boldsymbol{\Sigma}$ is a (pxp) matrix with $Var[X_i] = \sigma_{ii}$ and $\mathrm{Cov}[X_i, X_j] = \sigma_{ij}$, $i \neq j$

$|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$.

If $\sigma_{ij} = 0$, for all $i \neq j$, then $X_i, i = 1,2,\ldots,p$ are independent $N(\mu_i, \sigma_i^2)$ random variables.

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

For p =2, the bivariate normal $\boldsymbol{X \sim N_2(\mu, \Sigma)}$, where

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

has the following pdf

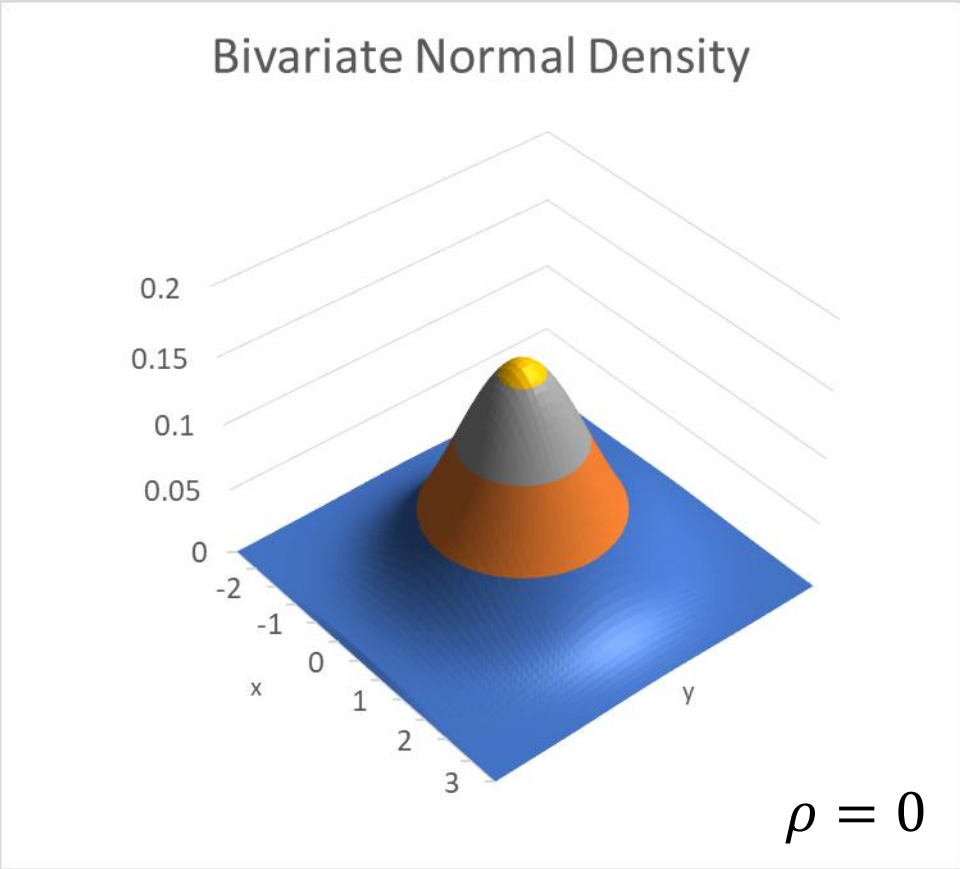
$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}\frac{1}{1-\rho^2}\left[\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right) + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2\right]\right\}$$

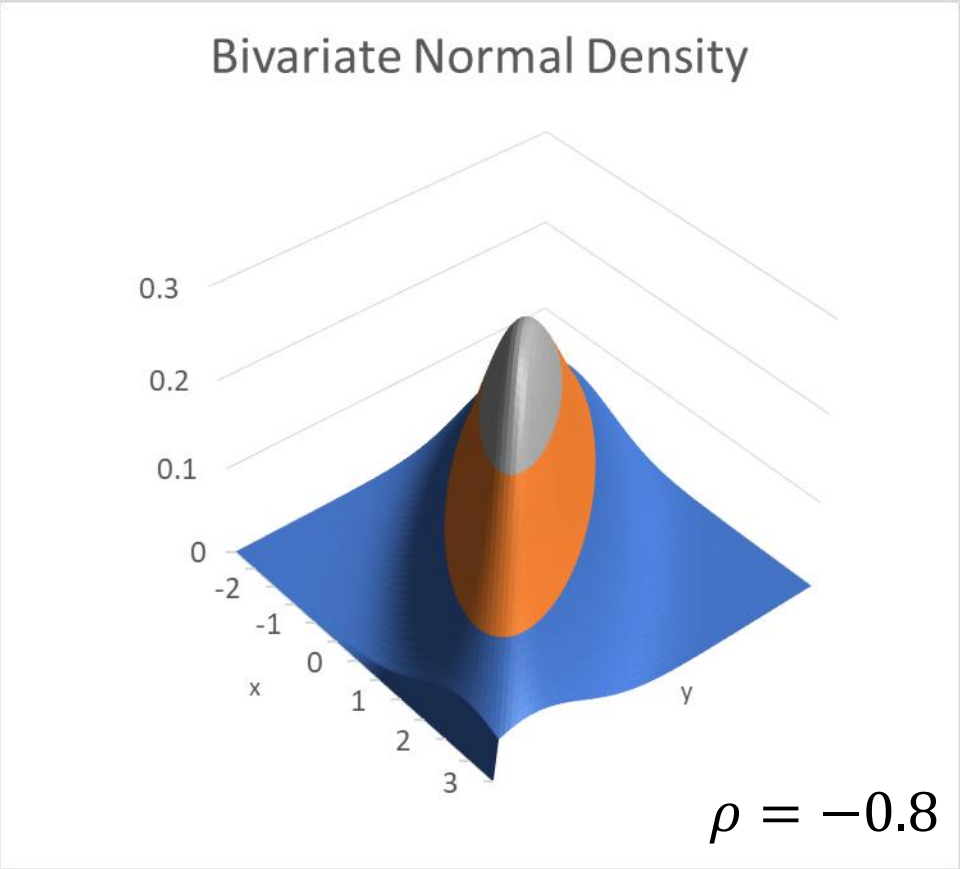
Letting $z_1 = \frac{x_1-\mu_1}{\sigma_1}$ and $z_2 = \frac{x_2-\mu_2}{\sigma_2}$ gives the *standardized bivariate normal* with pdf

$$f(z_1, z_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}\frac{1}{1-\rho^2}[z_1^2 - 2\rho z_1 z_2 + z_2^2]\right\}$$

THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.

Illustrations of the bivariate normal distribution





THIS CONTENT IS PROTECTED AND MAY NOT BE SHARED, UPLOADED, OR DISTRIBUTED.